25901154 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSECHEA00 - CHARACTERIZATION OF NANOMATERIALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **5** questions

Each question carries **4** marks

1. Differentiate between KLL and LMM emission in AES. [U] / [CO3]
2. Explain the mechanism of Internal Conversion (IC) and Intersystem Crossing (ISC) [U] / [CO3]
3. Calculate the concentration of a colloidal solution using the Beer-Lambert Law. Given its absorbance, path length, and extinction coefficient. [A] / [CO3]
4. Analyze the importance of combining TGA and DSC for complete thermal characterization of nanomaterials. [An] / [CO4]
5. How Zeta potential depends on pH? [U] / [CO5]
6. Write down the BET equation and identify each term involved. [K] / [CO4]
7. Recognise the scale does STM operate. [K] / [CO2]

[4x5 = 20]

Part B

Short Essay

Answer any **5** questions

Each question carries **7** marks

8. List and explain the advantages of Secondary Ion Mass Spectroscopy. [K] / [CO3]

9. What is a Supporting electrolyte in Polarography? [U] / [CO5]

10. Discuss the relevance of particle size distribution with respect to the size-dependent properties of quantum dots. [U] / [CO1]

11. Explain the working of a SEM with the aid of a labelled diagram. [K] / [CO1]

12. Explain the working principle of AFM. [U] / [CO2]

13. Explain the basic principle of Surface-Enhanced Raman Scattering (SERS). Discuss the two primary mechanisms responsible for the signal enhancement and mention two practical applications in nanotechnology. [An] / [CO3]

14. Discuss the principle of Small-Angle X-ray Scattering (SAXS). Explain how the "Guinier Plot" and the "Porod Law" are used to determine the size and surface characteristics of a nanoparticle. [U] / [CO4, CO6]

[7x5 = 35]

Part C

Essay

Answer any **1** questions

Each question carries **15** marks

15. Discuss the working principle of X-ray photoelectron spectroscopy. Also explain how is it useful for quantitative elemental analysis. [U] / [CO3]

16. Describe the basic intrumental features of FESEM with a schematic diagram. [U] / [CO1]

[15x1 = 15]